

CSA0404 – OPERATING SYSTEMS

EXPERIMENT 39

QUESTION

Develop a C program to simulate C-SCAN disk scheduling algorithm.

PROGRAM



The image shows a code editor with a C program for C-SCAN disk scheduling. The program prompts the user for the number of disk requests, the request queue, and the initial head position. It then calculates the total seek time and average seek time. The console output shows the user input and the program's results.

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int n, head, seek_time = 0;
6
7     printf("Enter the number of disk requests: ");
8     scanf("%d", &n);
9
10    int request_queue[n];
11
12    printf("Enter the disk request queue:\n");
13    for (int i = 0; i < n; i++) {
14        scanf("%d", &request_queue[i]);
15    }
16
17    printf("Enter the initial position of the disk head: ");
18    scanf("%d", &head);
19
20    for (int i = 0; i < n - 1; i++) {
21        for (int j = i + 1; j < n; j++) {
22            if (request_queue[i] > request_queue[j]) {
23                int temp = request_queue[i];
24                request_queue[i] = request_queue[j];
25                request_queue[j] = temp;
26            }
27        }
28    }
29
30    printf("\nC-SCAN Disk Scheduling:\n");
31
32    int index = 0;
```

Input

```
Enter the number of disk requests: 5
Enter the disk request queue:
50 10 20 70 30
Enter the initial position of the disk head: 40

C-SCAN Disk Scheduling:
Total Seek Time: 130
Average Seek Time: 26.00

...Program finished with exit code 0
Press ENTER to exit console.
```